3. Project Design Phase

3.1 System Design Overview

The Incident Lifecycle Automation solution is designed around the standard ServiceNow IT Service Management architecture.

The Incident module acts as the central component. Other ServiceNow components are connected to the incident lifecycle to provide service information, configuration information, knowledge assistance, change management, child incident management, and SLA tracking.

3.2 High-Level Architecture

The project can be represented using the following logical structure:

User

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Service Desk

↓

Incident Management

↓

Incident Classification

↓

Knowledge Management

↓

Level 2 Support

↓

Change Management

↓

Child Incident Management

↓

Incident Resolution

↓

Knowledge Creation

↓

Final Validation

3.3 Service Hierarchy

The service structure used in the project is:

Service

└── Remote Access

  └── Service Offering

    └── Corporate VPN

The incident is associated with this service hierarchy so that the affected business/IT service can be identified.

3.4 Incident Data Model

The primary incident record contains information such as:

Incident

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├── Caller

├── Short Description

├── Description

├── Channel

├── Category

├── Subcategory

├── Urgency

├── Service

├── Service Offering

├── Configuration Item

├── Assignment Group

├── Assigned To

├── State

├── Work Notes

├── Cause

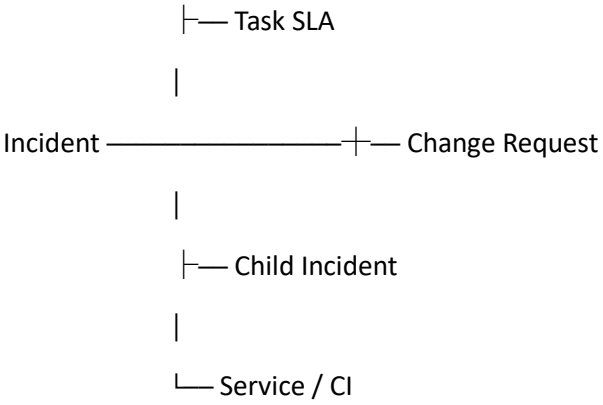
└── Resolution

3.5 Related Records

The incident acts as the central record connecting several supporting records.

┌── Knowledge Article

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This design provides a centralized view of the incident and its associated activities.

3.6 Incident Lifecycle Design

The project follows this lifecycle:

New

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Classified

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Service Desk Assignment

↓

Knowledge Assistance

↓

Network Assignment

↓

Level 2 Investigation

↓

Awaiting Change

↓

Investigation / Resolution

↓

Resolved

↓

Knowledge Creation

↓

Validation

The exact ServiceNow state transitions may depend on the platform configuration, but the project demonstrates the logical progression of the incident.

3.7 Knowledge Integration Design

Knowledge Management is integrated into the incident investigation process.

The design allows an agent to:

1. Open Agent Assist.
2. Search/view relevant knowledge.
3. Review troubleshooting instructions.
4. Mark the article as helpful.
5. Attach the article to the incident.
6. Record a comment about using the article.

This creates a relationship between the incident and the knowledge used to troubleshoot it.

3.8 Support Group Design

The support process uses multiple levels of support:

Service Desk

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| Escalation / Reassignment

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Network / Level 2 Support

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| Investigation

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Change Management (when required)

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Incident Resolution

This design establishes clear ownership as the incident moves between teams.

3.9 Child Incident Design

When multiple users experience the same issue, a child incident can be created from the parent incident.

Parent Incident

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├— Child Incident 1

├— Child Incident 2

└— Child Incident 3

This provides centralized tracking of a common issue while retaining individual user incidents.

3.10 Resolution Design

Resolution consists of three major activities:

1. Identify the probable cause.
2. Apply or document the solution.
3. Record resolution code and resolution notes.

For the demonstration:

Probable Cause: PowerEdge service was suspended and required restart.

Resolution Code: Workaround provided.

Resolution Notes: Restarted VPN-SRV-02 service as per emergency change request.

3.11 Design Principles

The project design follows these principles:

- Use standard ServiceNow functionality where possible.
- Maintain a clear incident lifecycle.
- Keep incident-related information centralized.
- Maintain relationships between related records.
- Provide clear ownership during escalation.
- Preserve resolution history.
- Support knowledge reuse.
- Maintain SLA visibility.